Cell Tuition
1-1 PRIVATE TUTORS

MEASUREMENT

Converting Units

Name _____ Date _____ No calculator Version 1

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KEY WORDS convert *(panoni)* · length *(roa)* · mass *(papatipu)* · capacity *(rōrahi)*

A Sensible units ●○○

Remember: Length: mm, cm, m, km. Mass: g, kg, t. Capacity: mL, L. Choose a unit that gives a number that is easy to picture.

1. Write the most sensible unit for measuring each of these.

a) The length of a pencil _____

b) The distance from Auckland to Wellington _____

c) The mass of an apple _____

d) The mass of a dog _____

e) The amount of water in a bath _____

f) The amount of medicine on a spoon _____

g) The thickness of a coin _____

h) The height of a house _____

B Metric prefixes ●○○

Remember: Kilo means 1,000. Centi means one hundredth. Milli means one thousandth. So 1 kilometre is 1,000 metres, and there are 100 centimetres and 1,000 millimetres in 1 metre.

2. Answer each question.

a) How many metres are there in 1 kilometre? _____

b) How many centimetres are there in 1 metre? _____

c) How many millimetres are there in 1 metre? _____

d) How many grams are there in 1 kilogram? _____

e) How many millilitres are there in 1 litre? _____

f) How many millimetres are there in 1 centimetre? _____

C Converting lengths ●●○

Remember: Multiply to change to a smaller unit (there will be more of them). Divide to change to a larger unit.

3. Write the missing number.

a) 4 cm = ___ mm

b) 3 m = ___ cm

c) 2 km = ___ m

d) 70 mm = ___ cm

e) 400 cm = ___ m

f) 5,000 m = ___ km

g) 2.5 m = ___ cm

h) 0.5 km = ___ m

i) 36 mm = ___ cm

j) 1.2 m = ___ mm

k) 75 cm = ___ m

l) 640 m = ___ km

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D Converting mass ●●○

> **Remember:** 1 g = 1,000 mg, 1 kg = 1,000 g, 1 tonne (t) = 1,000 kg.

4. Write the missing number.

a) 3 kg = ___ g **b)** 5,000 g = ___ kg **c)** 4 t = ___ kg **d)** 2,000 mg = ___ g

e) 1.5 kg = ___ g **f)** 750 g = ___ kg **g)** 2,500 kg = ___ t **h)** 6 g = ___ mg

E Converting capacity ●●○

> **Remember:** 1 L = 1,000 mL.

5. Write the missing number.

a) 2 L = ___ mL **b)** 4,000 mL = ___ L **c)** 0.5 L = ___ mL **d)** 3,200 mL = ___ L

e) 1.25 L = ___ mL **f)** 750 mL = ___ L **g)** 10 L = ___ mL **h)** 85 mL = ___ L

F Problems ●●●

6. A bottle holds 1.5 litres of juice. How many 250 mL glasses can be filled from it?

7. A piece of wood is 3.2 m long. It is cut into pieces that are 80 cm long. How many pieces are there?

8. A tin of beans has a mass of 450 g. Find the mass of 4 tins, in kilograms.

9. Mia is 1.2 m tall. Her little brother is 85 cm tall. How much taller is Mia, in centimetres?

★ Extension

10. Write the missing number. These take two steps.

a) 2 km = ___ cm **b)** 3 m = ___ mm **c)** 5,000,000 mg = ___ kg

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1-1 PRIVATE TUTORS

MEASUREMENT

Converting Units

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS convert *(panoni)* · length *(roa)* · mass *(papatipu)* · capacity *(rōrahi)*

A Sensible units ●○○

Remember: Length: mm, cm, m, km. Mass: g, kg, t. Capacity: mL, L. Choose a unit that gives a number that is easy to picture.

1. Write the most sensible unit for measuring each of these.

a) The width of a book _____ **b)** The distance from Hamilton to Taupo _____

c) The mass of an egg _____ **d)** The mass of a bicycle _____

e) The amount of petrol in a car _____ **f)** The amount of juice in a small glass _____

g) The thickness of a fingernail _____ **h)** The length of a bus _____

B Metric prefixes ●○○

Remember: Kilo means 1,000. Centi means one hundredth. Milli means one thousandth. So 1 kilometre is 1,000 metres, and there are 100 centimetres and 1,000 millimetres in 1 metre.

2. Answer each question.

a) How many grams are there in 1 kilogram? _____

b) How many millimetres are there in 1 centimetre? _____

c) How many metres are there in 1 kilometre? _____

d) How many millilitres are there in 1 litre? _____

e) How many centimetres are there in 1 metre? _____

f) How many kilograms are there in 1 tonne? _____

C Converting lengths ●●○

Remember: Multiply to change to a smaller unit (there will be more of them). Divide to change to a larger unit.

3. Write the missing number.

a) 6 cm = ___ mm **b)** 5 m = ___ cm **c)** 4 km = ___ m **d)** 90 mm = ___ cm

e) 200 cm = ___ m **f)** 3,000 m = ___ km **g)** 3.5 m = ___ cm **h)** 0.2 km = ___ m

i) 48 mm = ___ cm **j)** 1.5 m = ___ mm **k)** 45 cm = ___ m **l)** 820 m = ___ km

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D Converting mass ●●○

> **Remember:** 1 g = 1,000 mg, 1 kg = 1,000 g, 1 tonne (t) = 1,000 kg.

4. Write the missing number.

a) 4 kg = ___ g **b)** 7,000 g = ___ kg **c)** 6 t = ___ kg **d)** 3,000 mg = ___ g

e) 2.5 kg = ___ g **f)** 250 g = ___ kg **g)** 3,500 kg = ___ t **h)** 9 g = ___ mg

E Converting capacity ●●○

> **Remember:** 1 L = 1,000 mL.

5. Write the missing number.

a) 3 L = ___ mL **b)** 6,000 mL = ___ L **c)** 0.25 L = ___ mL **d)** 1,800 mL = ___ L

e) 2.75 L = ___ mL **f)** 250 mL = ___ L **g)** 20 L = ___ mL **h)** 65 mL = ___ L

F Problems ●●●

6. A bottle holds 2 litres of juice. How many 250 mL glasses can be filled from it?

7. A piece of wood is 2.4 m long. It is cut into pieces that are 60 cm long. How many pieces are there?

8. A tin of beans has a mass of 350 g. Find the mass of 6 tins, in kilograms.

9. Mia is 1.4 m tall. Her little brother is 75 cm tall. How much taller is Mia, in centimetres?

★ Extension

10. Write the missing number. These take two steps.

a) 3 km = ___ cm **b)** 2 m = ___ mm **c)** 4,000,000 mg = ___ kg

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